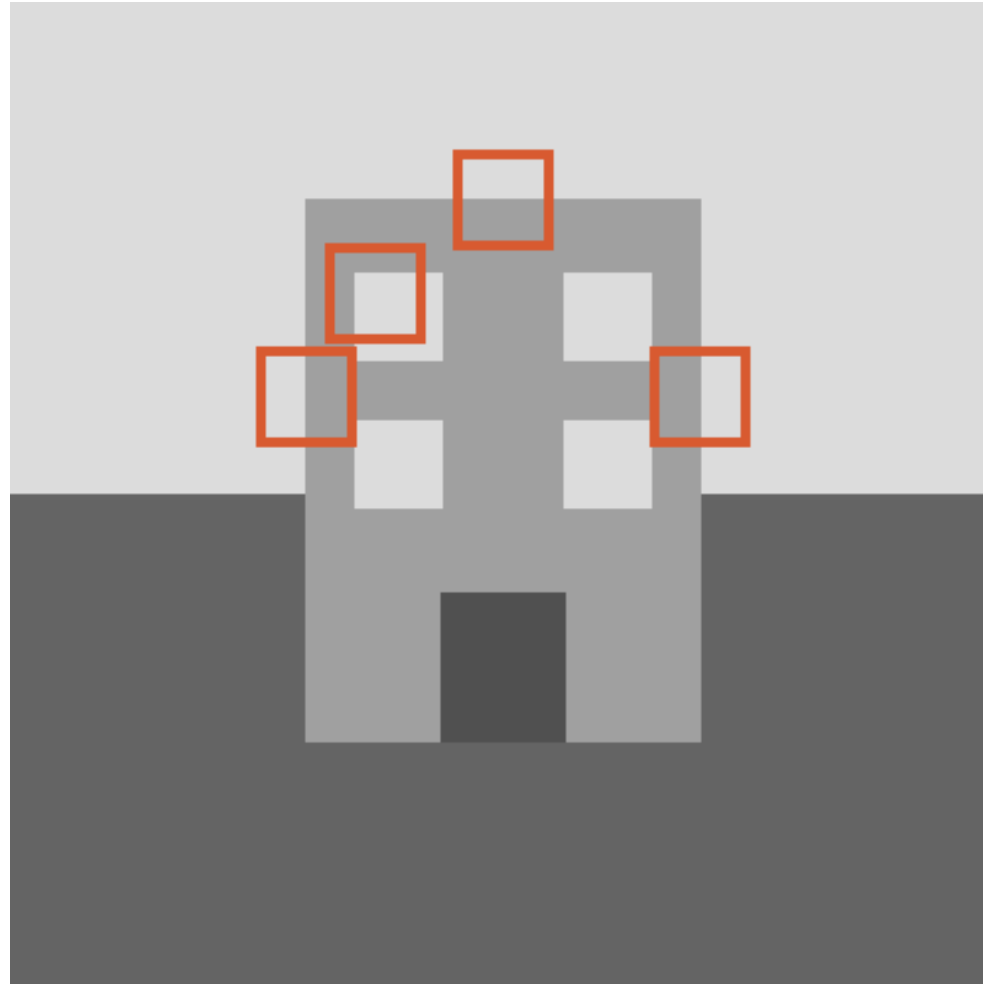


CONVOLUTIONAL & RECURRENT NEURAL NETWORKS

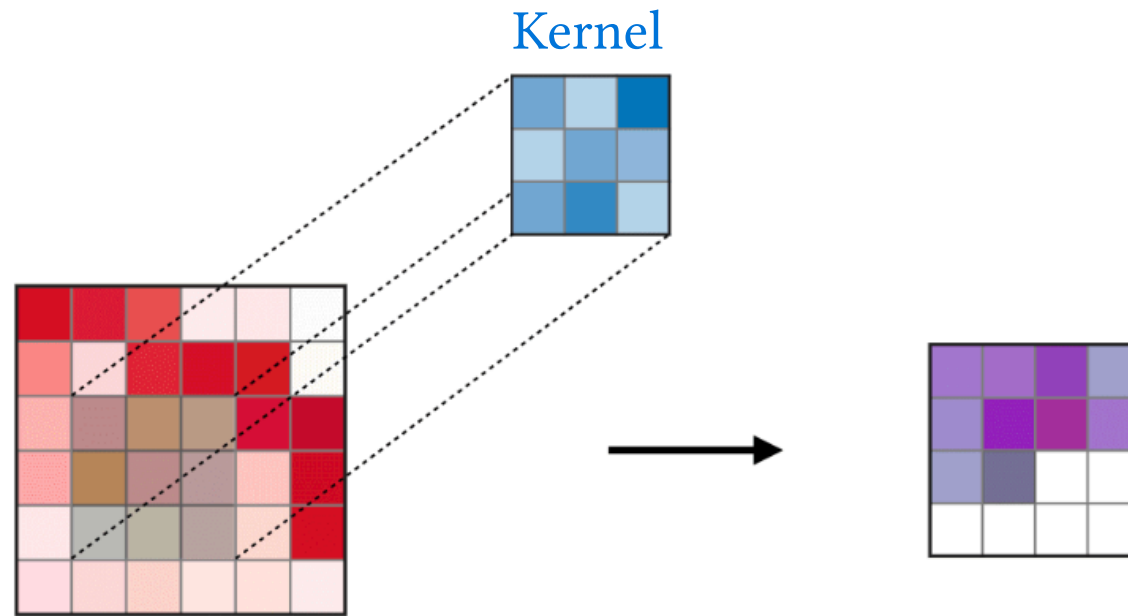
AI506 — Advanced Machine Learning

Simon Holm

May 2026



The Convolutional Layer:



The Convolutional Layer:

- 1D

$$(f * g)[n] = \sum_k f[k] \cdot g[n - k]$$

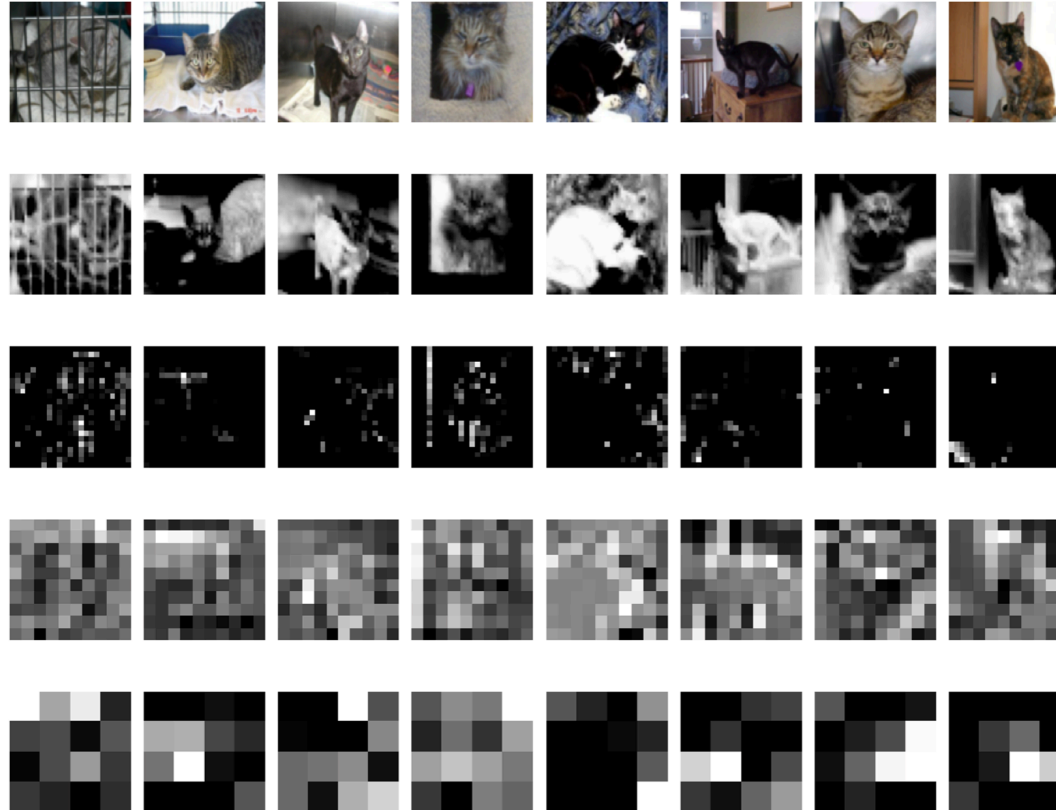
- 2D

$$(f * g)[n, m] = \sum_{k_1=0}^{k-1} \sum_{k_2=0}^{k-1} f[k_1, k_2] \cdot g[n - k_1, m - k_2]$$

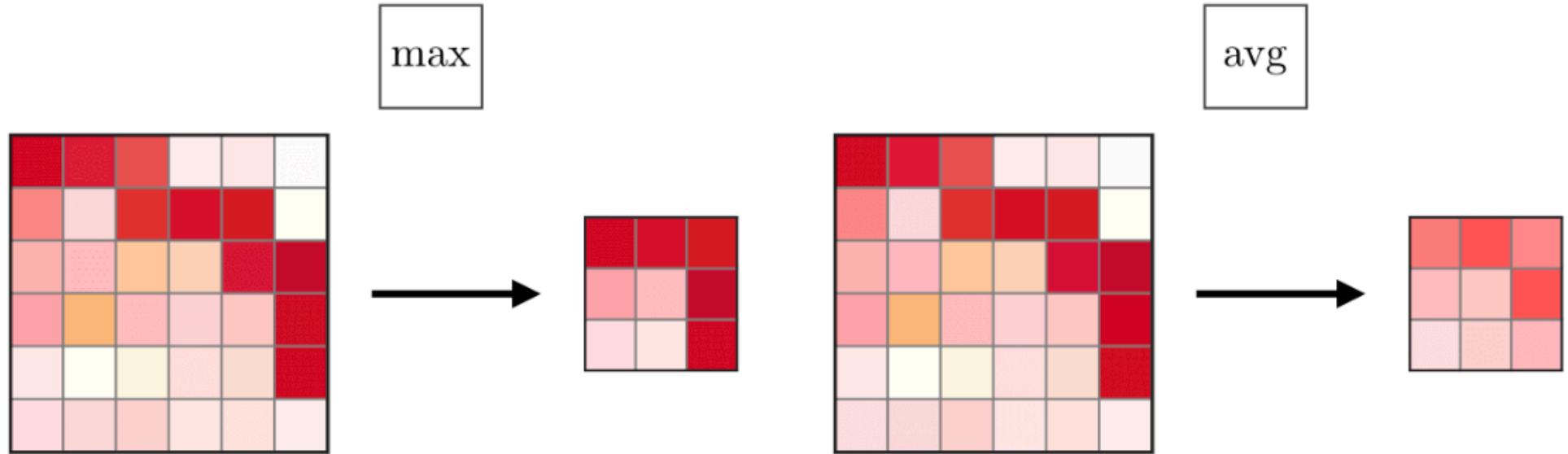
Where $k \in \mathbb{Z}^+$

$$k \ll n, m \Rightarrow |\mathbb{R}^{k \times k}| \ll |\mathbb{R}^{n \times m}|$$

From my project:



Pooling:



Let $\gamma = \|W_h\|$.

$$\gamma < 1 \implies \frac{\partial \mathcal{L}}{\partial h} \rightarrow 0 \quad (\text{vanishing}).$$

$$\gamma > 1 \implies \frac{\partial \mathcal{L}}{\partial h} \rightarrow \infty \quad (\text{exploding}).$$

$$h^{(l+1)} = \sigma(W \cdot h^{(l)} + b) + h^{(l)}$$

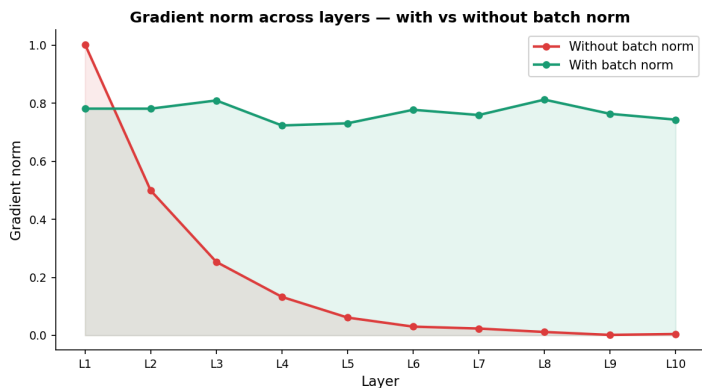
$$\frac{dh_{t+1}}{dh_0} = \frac{dh_{t+1}}{dh_t} + 1 \cdot \frac{dh_t}{dh_{t-1}} + 1 \cdot \dots \frac{dh_1}{dh_0} + 1 \geq 1$$

For gradients ≈ 0 ,

$$\nabla_{h^{(l)}} \mathcal{L} = \nabla_{h^{(l+1)}} \mathcal{L} \cdot 1 = \underbrace{\nabla_{h^{(l+1)}} \mathcal{L}}_{\text{unchanged}}$$

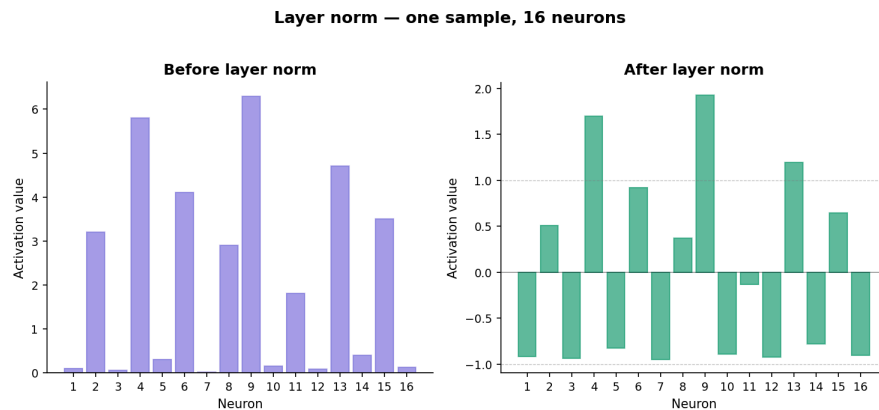
BatchNorm

$$\tilde{x}_i^k = \frac{x_i^k - \mu_B^k}{\sqrt{(\sigma_B^k)^2 + \varepsilon}}$$

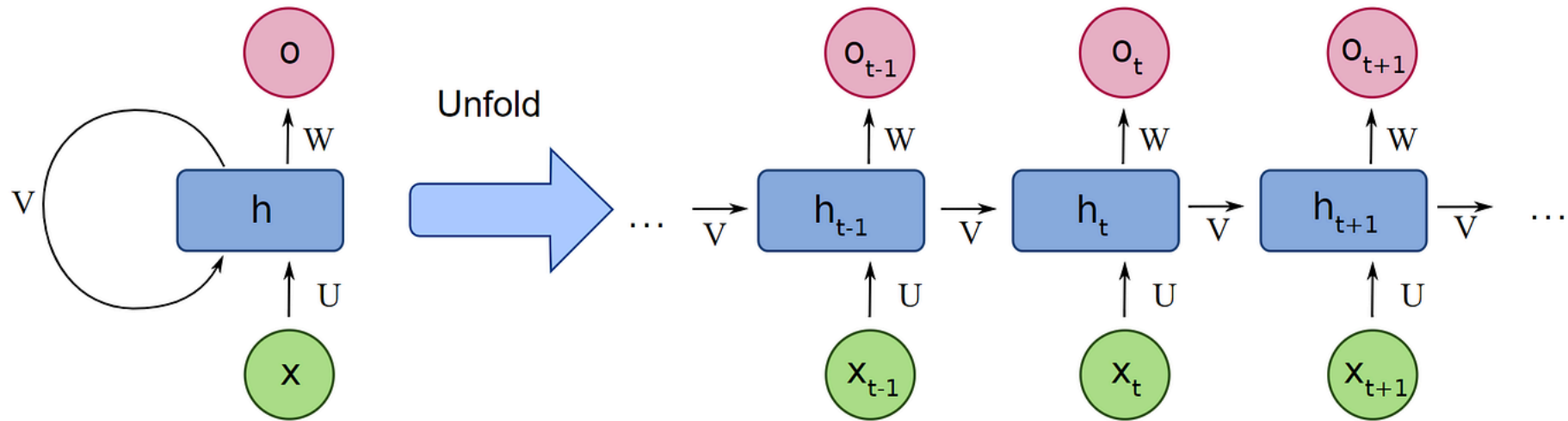


LayerNorm

$$\tilde{x}_i = \frac{x_i - \mu_s}{\sqrt{(\sigma_s)^2 + \varepsilon}}$$



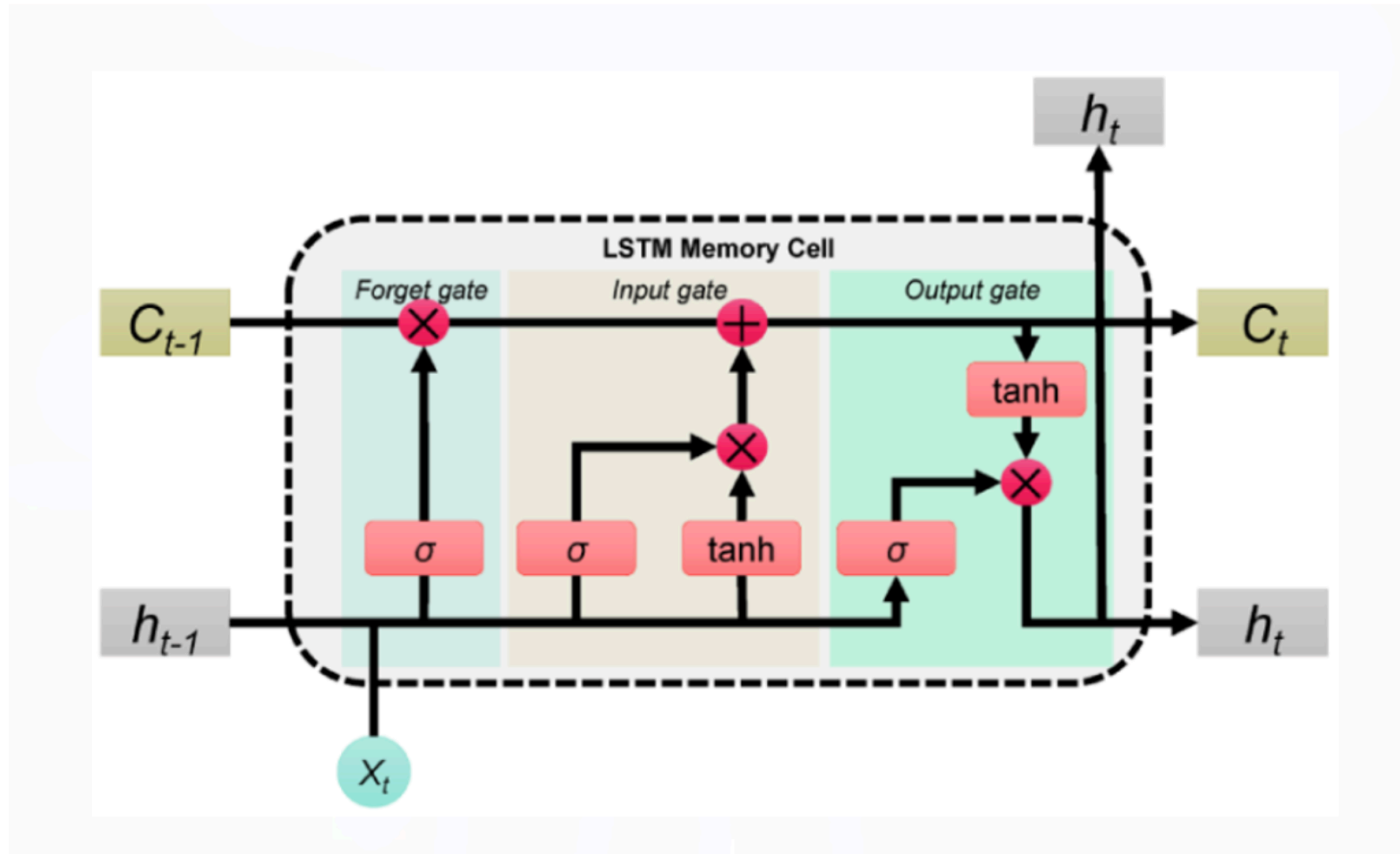
$$h_t = \tanh(W_{hh} \cdot h_{t-1} + W_{xh} \cdot x_t + b).$$



Recall that for $\gamma = \|W_h\|$.

$$\gamma < 1 \implies \frac{\partial \mathcal{L}}{\partial h} \rightarrow 0 \quad (\text{vanishing}).$$

The cat, which was sitting on the mat near the window in the cold room, **was hungry**.



From my project:

